

Recommendations — hgsoc-tuboovarian-stage3c-r0-hrd-pen ding-idyq

Ranked therapeutic options with likelihood, toxicity, mechanism risk, and key references

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Recommendations — hgsoc-tuboovarian-stage3c-r0-hrd-pending

Profile snapshot

- **Primary site:** ovary
- **Histology:** high-grade serous carcinoma (tubo-ovarian / primary peritoneal; WT1+, PAX8+, SOX17+, CK7+)
- **Stage:** IIIc
- **ECOG:** 1
- **Age band:** 70-79
- **Sex:** female
- **Biomarkers:** TP53 V157G (somatic) somatic missense mutation in the DNA-binding domain; concordant with diffuse p53 IHC overexpression; BRCA1/2 (germline and somatic) wild-type — no pathogenic variant germline (74-gene panel) and no actionable somatic alteration on tumor NGS (BRCA-proficient); HRD / genomic instability (tumor) genomic instability PRESENT but qualitative only — multiple chromosomal imbalances / frequent LOH; BRCA-proficient; NO quantitative GIS/HRD score reported; MSI / MMR Microsatellite Stable (MSS); TMB Low (1 mut/Mb); PIK3CA amplification (somatic) focal amplification; MAP2K4 deletion (somatic) focal deletion; ER positive (~80% of tumor nuclei, weak-moderate intensity); PR negative; HER2 HER2-low (IHC 1+); FOLR1 (folate receptor alpha) negative (<75% of cells at moderate/strong membranous intensity); CA-125 normalized on therapy (peak 940 -> 11.6 U/mL); Signatera ctDNA (MRD) cleared to Not Detected (30.3 -> 3.32 -> 0 MTM/mL)
- **Efficacy/toxicity weight:** 0.7
- **Toxicity vetoes:** additional peripheral neuropathy, severe myelosuppression / febrile neutropenia
- **Modality constraints:** oral or low-infusion-burden preferred, caution with agents carrying thrombotic or bleeding risk (history of high VTE risk; prior IVC filter)

8 ranked therapeutic options across 5 targetable features. Biomarker workup steps live in the standalone Target validation paths report.

Hrd Genomic Instability interventions

1. Rucaparib (1L maintenance)

Likelihood of effect: Moderate-to-high for PFS: replicated all-comer phase 3 (ATHENA-MONO ITT HR 0.52, 20.2 vs 9.2 mo) qualifying a BRCA-wild-type tumor without a BRCA or GIS gate.

Toxicity burden: Moderate (grade 3+ anemia 28.7%, neutropenia 14.6%, transaminase rise 10.6%)

Counter-productive MoA:Low (PARP myelosuppression is a patient-AE burden, not a mechanism that blunts the maintenance goal in genomically-unstable HGSC.)

Overall: Replicated all-comer PARP maintenance that qualifies a BRCA-wild-type tumor with no GIS gate; near-interchangeable with niraparib but lacks its tumor-specific pharmacology, and carries the same myelosuppression veto.

Key references: PMID 35658487 · NCT03522246

2. Niraparib (1L maintenance)

Likelihood of effect: High for PFS as an evidence fit: all-comer PRIMA phase 3 (HR 0.62, 13.8 vs 8.2 mo) plus BRCA-wild-type PDX pharmacology (pmid:30647846); deliverability gated by marrow reserve.

Toxicity burden: Moderate (grade 3+ thrombocytopenia 28.7%, anemia 31%, neutropenia 12.8%)

Counter-productive MoA: **Moderate** (Cytopenia-driven dose interruption on thin baseline marrow reserve can erode maintenance dose intensity; the advocate and conservative dissented on this toxicity-as-deliverability axis.)

Overall: Cleanest all-comer guideline and evidence fit with the only tumor-specific BRCA-wild-type pharmacology, but its grade 3+ thrombocytopenia/anemia hit the patient's named veto; deliverable only behind a documented hematology dose plan.

Key references: PMID 31562799 · PMID 30647846 · NCT02655016

3. Olaparib + bevacizumab (PAOLA-1 maintenance, contingent on validated HRD-positive)

Likelihood of effect: High ONLY if GIS ≥ 42 confirms HRD-positive: PAOLA-1 subgroup HR 0.33 (37.2 vs 17.7 mo); contingent and foreclosed below 42, ITT effect is a smaller HR 0.59.

Toxicity burden: High (combination grade 3+ AE ~57%; hypertension 19%, anemia 17%, up to 20% olaparib discontinuation)

Counter-productive MoA: **High** (Anti-angiogenic VTE/bleeding stacked on PARP myelosuppression against an IVC-filter history; the conservative's toxicity veto stood on this mechanism.)

Overall: Highest-magnitude maintenance if HRD-positive is confirmed, but contingent on a GIS she lacks and held back by a toxicity veto and an evidence-quality dissent on a subgroup-only effect estimate.

Key references: PMID 31851799 · NCT02477644

Evidence in detail

Rucaparib (1L maintenance)

****Monk et al. J Clin Oncol 2022****

- **Disease context:** newly diagnosed advanced high-grade epithelial ovarian cancer, response after 1L platinum doublet (maintenance) — All-comers (4:1 randomization), HRD-positive and HRD-negative strata. Benefit held in the ITT population and in the non-nested HRD-negative subgroup, so a BRCA-WT, HRD-unquantified tumor still qualifies.
- **n:** 538
- **Toxicities:** anemia (grade ≥ 3): 28.7%; neutropenia (grade ≥ 3): 14.6%; ALT/AST elevation (grade ≥ 3): 10.6%; treatment discontinuation (AE) (grade any): 11.8%
- **Efficacy:** HR_PFS 0.52 HR (95% CI 0.4-0.68); HR_PFS 0.47 HR (95% CI 0.31-0.72); PFS 20.2 months

Niraparib (1L maintenance)

****Gonzalez-Martin et al. *N Engl J Med* 2019****

- **Disease context:** newly diagnosed advanced high-grade serous/endometrioid ovarian cancer, response after 1L platinum (maintenance) — All-comers, both HRD-positive and HRD-negative strata; no BRCA mutation and no validated HRD score required for enrolment. The all-comer enrolment is what makes PRIMA the cleanest fit for this BRCA-proficient, genomically-unstable tumor.
- **n:** 733
- **Toxicities:** anemia (grade ≥ 3): 31.0%; thrombocytopenia (grade ≥ 3): 28.7%; neutropenia (grade ≥ 3): 12.8%
- **Efficacy:** HR_PFS 0.62 HR (95% CI 0.5-0.76); HR_PFS 0.43 HR (95% CI 0.31-0.59); PFS 13.8 months; HR_OS 0.7 HR (95% CI 0.44-1.11)

Olaparib + bevacizumab (PAOLA-1 maintenance, contingent on validated HRD-positive)

****Ray-Coquard et al. *N Engl J Med* 2019****

- **Disease context:** newly diagnosed advanced high-grade serous ovarian cancer receiving 1L bevacizumab (maintenance) — All-comers enrolled, but the predefined and label-relevant benefit is in the HRD-positive subgroup (Myriad MyChoice GIS ≥ 42 or tumor BRCA mutation). This patient is BRCA wild-type and has only a qualitative Altera instability call, so eligibility is CONTINGENT on a validated GIS she does not yet have.
- **n:** 806
- **Toxicities:** hypertension (grade ≥ 3): 19%; anemia (grade ≥ 3): 17%; lymphopenia (grade ≥ 3): 7%; fatigue/asthenia (grade ≥ 3): 5%
- **Efficacy:** HR_PFS 0.59 HR (95% CI 0.49-0.72); HR_PFS 0.33 HR (95% CI 0.25-0.45); HR_PFS 0.43 HR (95% CI 0.28-0.66)

Vegf Angiogenesis interventions

1. Bevacizumab (continued maintenance)

Likelihood of effect: Moderate for PFS only: GOG-218 HR 0.717 (14.1 vs 10.3 mo), RoB2-low but no overall-population OS gain; smallest maintenance effect on the board.

Toxicity burden: Low (grade 2+ hypertension 22.9%, GI-wall disruption 2.6%; VTE/bleeding class caution)

Counter-productive MoA: Low (Anti-angiogenic wound-healing and VTE/bleeding risk is a patient-AE caution here, not a mechanism that blunts maintenance benefit; risktaker dissent was magnitude/preference-flavored.)

Overall: The only maintenance that clears both named vetoes on a backbone she already tolerates, but PFS-only with no OS gain and a live VTE/bleeding caution against her IVC-filter history.

Key references: PMID 22204724 · PMID 22204725 · NCT00262847

Evidence in detail

Bevacizumab (continued maintenance)

****Burger et al. *N Engl J Med* 2011****

- **Disease context:** newly diagnosed stage III-IV epithelial ovarian / primary peritoneal cancer (1L) — All-comers, no biomarker selection. Matches the VEGF/angiogenesis axis the patient is already being treated on.
- **n:** 1873
- **Toxicities:** hypertension (grade ≥ 2): 22.9%; **GI perforation/wall disruption** (grade any): 2.6%
- **Efficacy:** HR_PFS 0.717 HR (95% CI 0.625-0.824); **PFS** 14.1 months

****Perren et al. *N Engl J Med* 2011****

- **Disease context:** newly diagnosed epithelial ovarian / primary peritoneal cancer (1L) — All-comers, no biomarker gate. OS benefit concentrated in the high-risk subset (stage IV or suboptimally debulked); this stage IIIc R0 patient sits adjacent to that subset.
- **n:** 1528
- **Toxicities:** hypertension (grade ≥ 2): 18%
- **Efficacy:** HR_PFS 0.81 HR (95% CI 0.7-0.94); **OS** 36.6 months

II12 Tme interventions

1. GEN-1 / IMNN-001 (IL-12 immunogene, OVATION 2)

Likelihood of effect: Moderate but immature: OVATION 2 ITT PFS HR 0.79 (14.9 vs 11.9 mo) and OS HR 0.69 from sponsor topline; CIs unreported, regimen-matched to this patient.

Toxicity burden: Low (IP delivery reported well tolerated; no per-term grade 3+ table published)

Counter-productive MoA: Low (IL-12 immunogene acts via the peritoneal TME, sidestepping the foreclosed MSS/TMB-low checkpoint axis; no plausible counter-productive vector flagged.)

Overall: On-study IL-12 continuation that sidesteps the foreclosed checkpoint axis and carries a real OS signal, but efficacy is sponsor-topline-immature and delivery is intraperitoneal, not oral.

Key references: NCT03393884 · PMID 20033066

Er Expression interventions

1. Letrozole (endocrine maintenance)

Likelihood of effect: Low and weakly evidenced: only a retrospective ROBINS-I-serious series (60% vs 38.5% RFS at 24 mo, pmid:29157627); PR-negativity tempers the ER-driven signal.

Toxicity burden: Low (arthralgia, bone-density loss on prolonged aromatase inhibition; no named veto triggered)

Counter-productive MoA: Low (Endocrine blockade has no plausible counter-productive vector in ER-positive HGSC; the dissent here is evidence-quality, not mechanism.)

Overall: The lowest-toxicity oral option that clears every veto and fits the age band, but the evidence is a

single retrospective ROBINS-I-serious series and PR-negativity tempers the expected magnitude.

Key references: PMID 29157627

Evidence in detail

Letrozole (endocrine maintenance)

****Heinzelmann-Schwarz et al. *Gynecol Oncol* 2018****

- **Disease context:**ER-positive high-grade serous ovarian cancer, maintenance / later-line (maintenance) — ER-positive HGSOC by IHC; a translational study with a clinical letrozole-maintenance series, enriched for platinum-resistant and residual-disease patients. The patient is ER-positive (~80% nuclei) but PR-negative, which tempers expected magnitude.
- **n:** —
- **Toxicities:** No per-term toxicity table in the source; aromatase-inhibitor class toxicity is arthralgia and bone-density loss on prolonged use. Oral, low-burden, no named patient veto applies.
- **Efficacy:**RFS 60%; RFS 87.5%

Her2 Low interventions

1. Trastuzumab deruxtecan (T-DXd), HER2-low (IHC 1+), investigational

Likelihood of effect: Low-to-moderate, cross-tumor only: no ovarian IHC 1+ data; rests on HER2-low breast extrapolation (DESTINY-Breast04 PFS HR 0.51) plus the DXd bystander mechanism.

Toxicity burden: High (ILD/pneumonitis 10-12% incl. fatal cases; grade 3+ AE 52.6%; nausea; anemia)

Counter-productive MoA:Moderate (Bystander DXd reach is unproven at IHC 1+ antigen density; sparse HER2-positive cells may not seed enough payload to kill neighbors.)

Overall:The one HER2-directed option that clears both named vetoes, but investigational at IHC 1+ ovarian on breast-only extrapolation — a relapse consideration after a confirmatory re-stain, not a maintenance pick.

Key references: PMID 35665782 · PMID 37870536 · PMID 39282896

2. Disitamab vedotin (RC48), HER2-directed ADC

Likelihood of effect: Not assessable for benefit here: the only HER2-low read is an abstract-only PAM-selected breast series (ORR ~34%); no ovarian data.

Toxicity burden: High (peripheral neuropathy 18.7% G3+, neutropenia 12.1% G3+, grade 3+ TRAE 54.2%, leukopenia 50.5%)

Counter-productive MoA:High (MMAE payload drives peripheral neuropathy and myelosuppression that hit both named vetoes head-on; the unanimous veto stood on this toxicity-as-mechanism.)

Overall:Board-vetoed by all five personas: the MMAE payload triggers both the peripheral-neuropathy and myelosuppression vetoes, with only abstract-only HER2-low breast efficacy and no ovarian data.

Key references: NCT06003231 · PMID 37988648

Evidence in detail

Trastuzumab deruxtecan (T-DXd), HER2-low (IHC 1+), investigational

****Modi et al. *N Engl J Med* 2022****

- **Disease context:**previously treated HER2-low (IHC 1+ or 2+/ISH-) metastatic breast cancer (2L+) — HER2-low breast cancer; the hormone-receptor-positive subgroup mirrors this patient's ER-positive status. Direct evidence that an IHC 1+ tumor responds to T-DXd, though the disease is breast rather than ovarian.
- **n:** 557
- **Toxicities:**adverse event (grade ≥ 3): 52.6%; **ILD/pneumonitis** (grade any): 12.1%; **ILD/pneumonitis** (grade 5): 0.8%; **nausea** (grade any): 75.7% (n=371); **anemia** (grade any): 36.9% (n=371); **neutrophil count decreased** (grade any): 21.8% (n=371); **fatigue** (grade any): 29.6% (n=371)
- **Efficacy:**HR_PFS 0.51 HR (95% CI 0.4-0.64); **HR_OS** 0.64 HR; **HR_PFS** 0.5 HR; **HR_OS** 0.64 HR

****Meric-Bernstam et al. *J Clin Oncol* 2024****

- **Disease context:**previously treated HER2-expressing advanced ovarian cancer (DESTINY-PanTumor02 ovarian cohort) (2L+) — Ovarian cohort enrolled at HER2 IHC 2+ or 3+. This patient is IHC 1+, below that entry bar, so the ovarian activity reported here comes from a higher-expression population than hers.
- **n:** 40
- **Toxicities:**treatment-related AE (grade ≥ 3): 40.8%; **ILD/pneumonitis** (grade any): 10.5%
- **Efficacy:**ORR 45.0% (95% CI 29.3-61.5); **DoR** 11.3 months (95% CI 4.1-22.1); **PFS** 5.9 months; **OS** 13.2 months; **ORR** 61.3% (95% CI 49.4-72.4); **TRAE_rate** 40.8%

****Bardia et al. *N Engl J Med* 2024****

- **Disease context:**HR-positive HER2-low or HER2-ultralow metastatic breast cancer after endocrine therapy (2L+) — HR-positive disease spanning HER2-low and HER2-ultralow (IHC 0 with faint membrane staining). The ultralow stratum is the closest expression analog to her IHC 1+, ER-positive tumor.
- **n:** 866
- **Toxicities:**adverse event (grade ≥ 3): 52.8%; **ILD/pneumonitis** (grade any): 11.3%; **ILD/pneumonitis** (grade 5)
- **Efficacy:**HR_PFS 0.62 HR (95% CI 0.52-0.75); **PFS** 13.2 months; **HR_PFS**

Disitamab vedotin (RC48), HER2-directed ADC

****Sheng et al. *J Clin Oncol* 2024****

- **Disease context:**HER2-positive (IHC 2+/3+) locally advanced or metastatic urothelial carcinoma (2L+) — HER2-positive urothelial carcinoma pooled from two phase 2 trials. Different tumor type and a higher HER2 tier than hers, but the best-powered published read on the drug's MMAE toxicity, which is what her neuropathy veto turns on.
- **n:** 107
- **Toxicities:**treatment-related AE (grade ≥ 3): 54.2%; **peripheral sensory neuropathy** (grade any): 68.2%;

peripheral sensory neuropathy (grade ≥ 3): 18.7%; **neutropenia** (grade any): 42.1%; **neutropenia** (grade ≥ 3): 12.1%; **leukopenia** (grade any): 50.5%

- **Efficacy:ORR** 50.5% (95% CI 40.6-60.3); **PFS** 5.9 months (95% CI 4.3-7.2); **OS** 14.2 months (95% CI 9.7-18.8); **DoR** 7.3 months (95% CI 5.7-10.8); **TRAE_rate** 54.2%